

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Cloud and Orchestrate”

Project Title – [Library AI Agent]

Domain of Project – [Education]

Student Name	Recent Passport Photo	Email ID	Mobile number [WhatsApp]
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Problem Statement -

Problem Statement : Library AI Agent

The Challenge :

Students often spend a lot of time searching for the right books and study materials in the library. They may not know which books are best for their subject or whether a book is available. Information about books, syllabi, and library resources is scattered, making it difficult to find the right learning material quickly.

The Objective :

Build an AI-powered Library Agent that helps students quickly find the best learning resources according to their academic needs.

The solution should provide:

- **Smart Book Recommendation**

Recommend books based on the student's subject, semester, course, or exam preparation.

- **Library Knowledge Retrieval (RAG)**

Search the library knowledge base and provide relevant books, reading lists, syllabi, and department resources using AI.

- **Book Search & Availability**

Help students search for books by title, author, or subject and show whether the book is available in the library.

- **Syllabus & Learning Support**

Answer questions related to course syllabi, recommended books, and department resources.

- **AI-Powered Library Assistant**

Understand student questions written in natural language and provide accurate book recommendations using IBM Granite and RAG.

- **Easy-to-Use Chat Interface**

Provide a simple chatbot interface where students can ask questions in normal language and instantly receive helpful responses.

Proposed Solution -

Proposed Solution: AI-Powered Library Agent

The proposed solution is an **AI-powered Library Agent** built using **IBM watsonx.ai, IBM Granite Model, Flask, and RAG (Retrieval-Augmented Generation)**. It helps students quickly find books, study materials, and library resources through a smart chatbot.

The system uses a **RAG pipeline** to search the library knowledge base, which contains the **book catalog, reading lists, course syllabi, and department guides**. The information is converted into embeddings and stored in a **ChromaDB vector database** for fast and accurate retrieval.

When a student asks a question, the chatbot understands the query using **IBM Granite** and searches the knowledge base to provide the most relevant answer.

The Library AI Agent provides the following features:

Book Search – Find books by title, author, or subject.

Book Recommendation – Suggest books based on the student's subject, course, or exam preparation.

Library Knowledge Retrieval – Retrieve information from the library catalog, syllabi, reading lists, and department guides using RAG.

Syllabus Support – Help students find recommended books and learning resources for their courses.

Easy Chat Interface – Allow students to ask questions in simple language and receive quick and accurate responses.

The chatbot is built using **Flask** with a simple web interface created using **HTML, CSS, and JavaScript**. **IBM watsonx.ai** and the **Granite model** are used to understand user queries and generate intelligent responses.

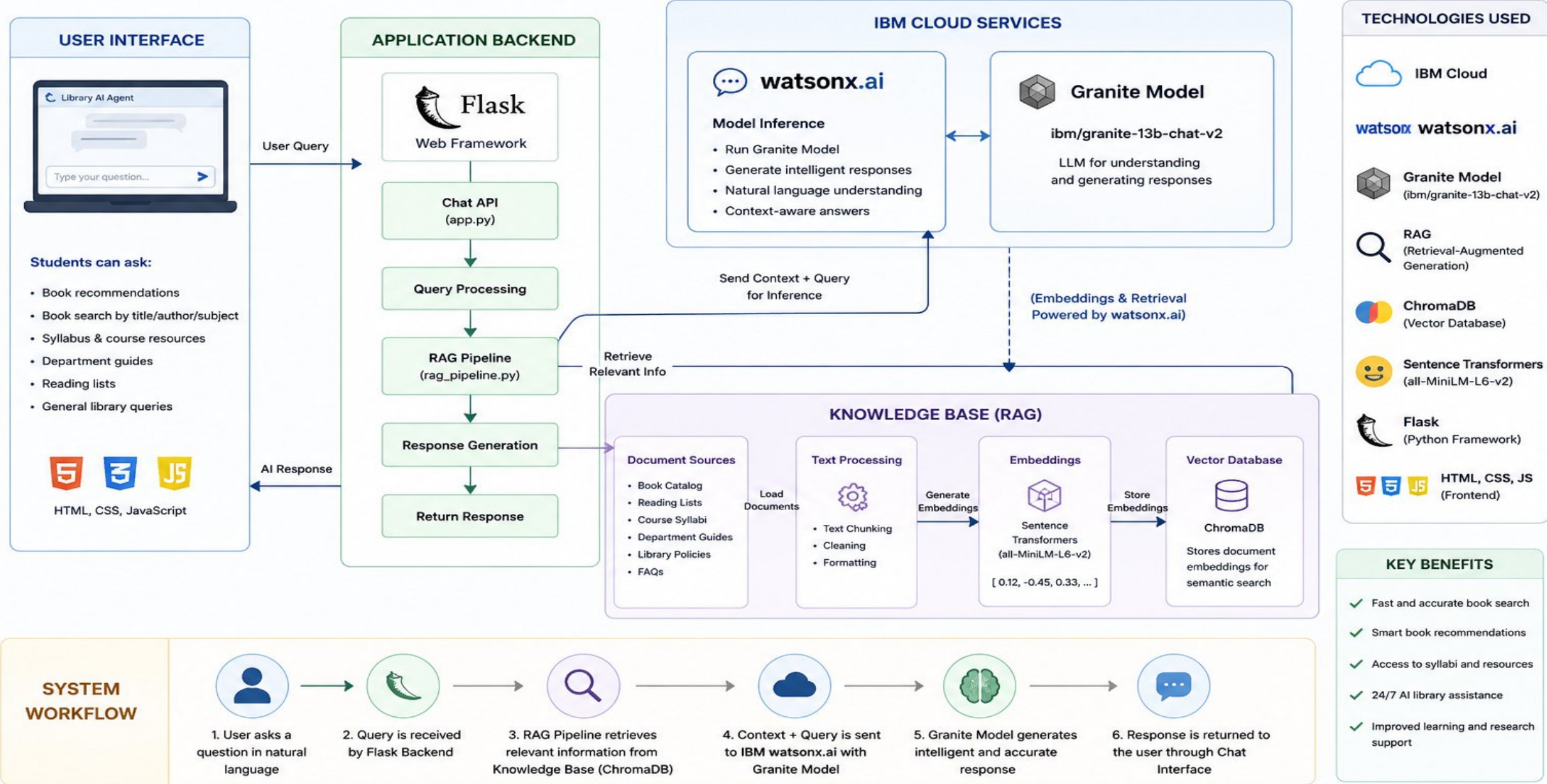
Overall, this project provides a **smart, fast, and user-friendly library assistant** that helps students easily find learning resources, saves time, and improves access to library materials.

Technology Used

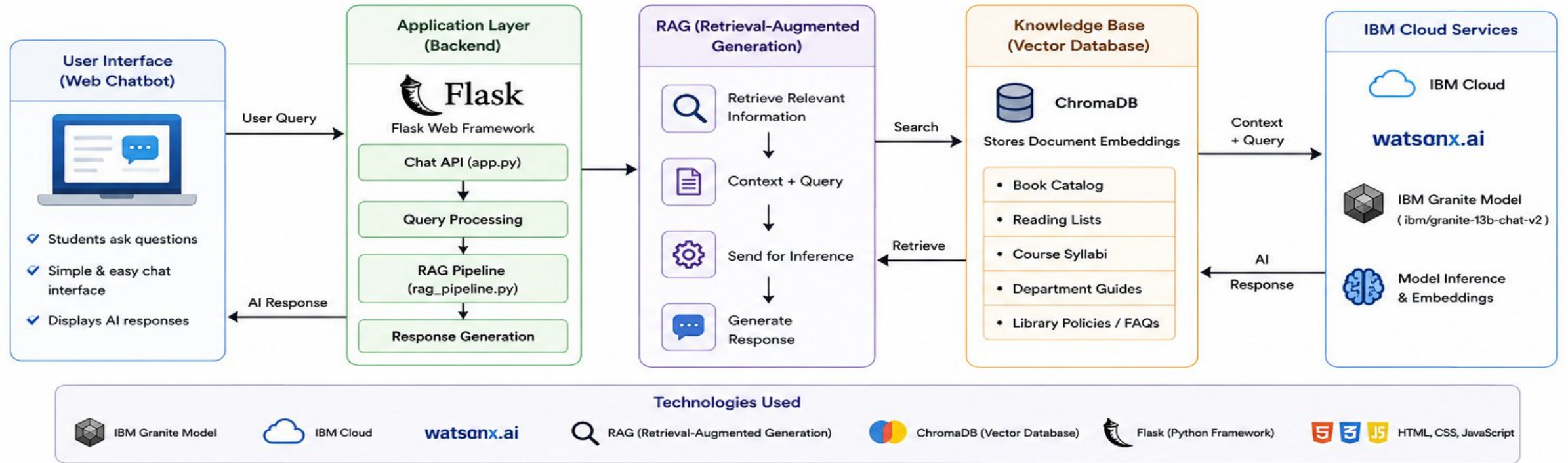
- **IBM Granite Model (ibm/granite-13b-chat-v2)** – Understands user questions and generates intelligent responses.
- **IBM Cloud** – Provides a secure platform to run the Library AI Agent.
- **IBM watsonx.ai** – Used for AI model integration and response generation.
- **RAG (Retrieval-Augmented Generation)** – Retrieves relevant library information before generating answers.
- **ChromaDB (Vector Database)** – Stores document embeddings for fast and accurate search.
- **Flask** – Backend framework for the chatbot.
- **HTML, CSS & JavaScript** – Used to build the chatbot's web interface.

Architecture Blueprint – Library AI Agent

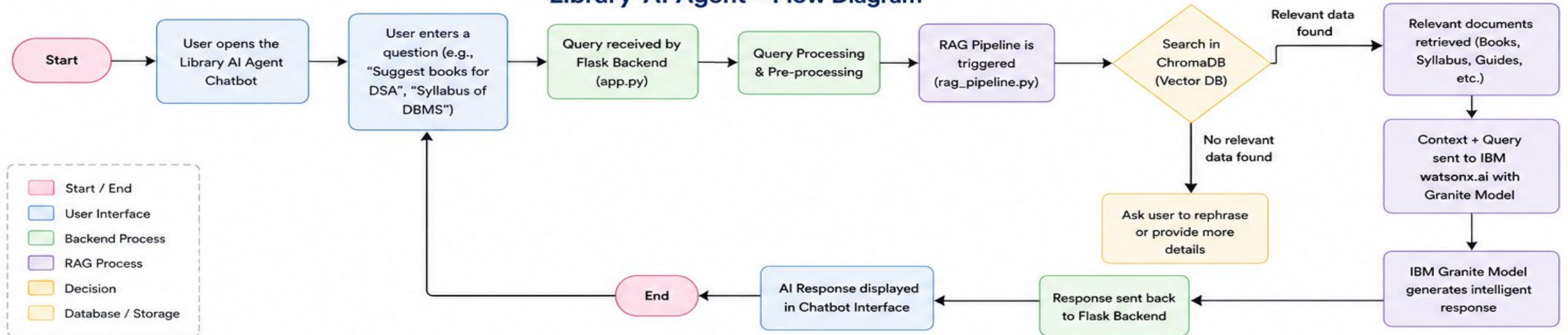
AI-Powered Library Assistant using IBM Cloud, watsonx.ai and RAG



Library AI Agent – Block Diagram



Library AI Agent – Flow Diagram



Role of Agentic AI in the solution

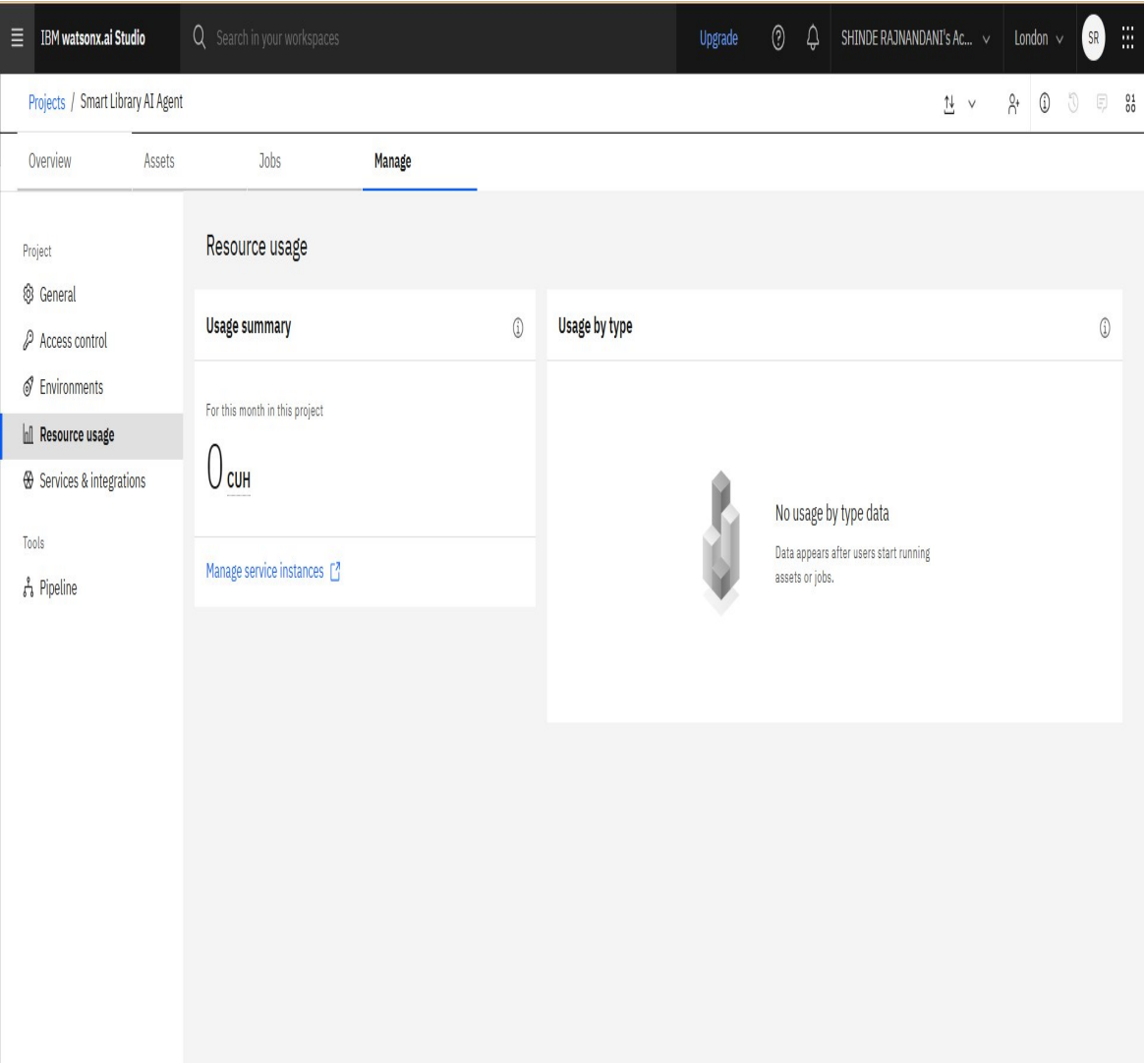
Agentic AI acts as the intelligent core of the **Library AI Agent**, where different AI functions work together to help students find the right learning resources quickly and accurately.

- **Book Recommendation Agent** recommends books based on the student's subject, semester, course, or exam preparation (such as GATE).
- **Library Knowledge Agent** retrieves relevant information from the library knowledge base using the **RAG (Retrieval-Augmented Generation)** approach, including books, syllabi, reading lists, and department resources.
- **Availability & Search Agent** checks book availability and helps students search books by title, author, or subject.
- **AI Assistant Agent** understands questions written in natural language and generates accurate, personalized responses using **IBM Granite** and **watsonx.ai**.

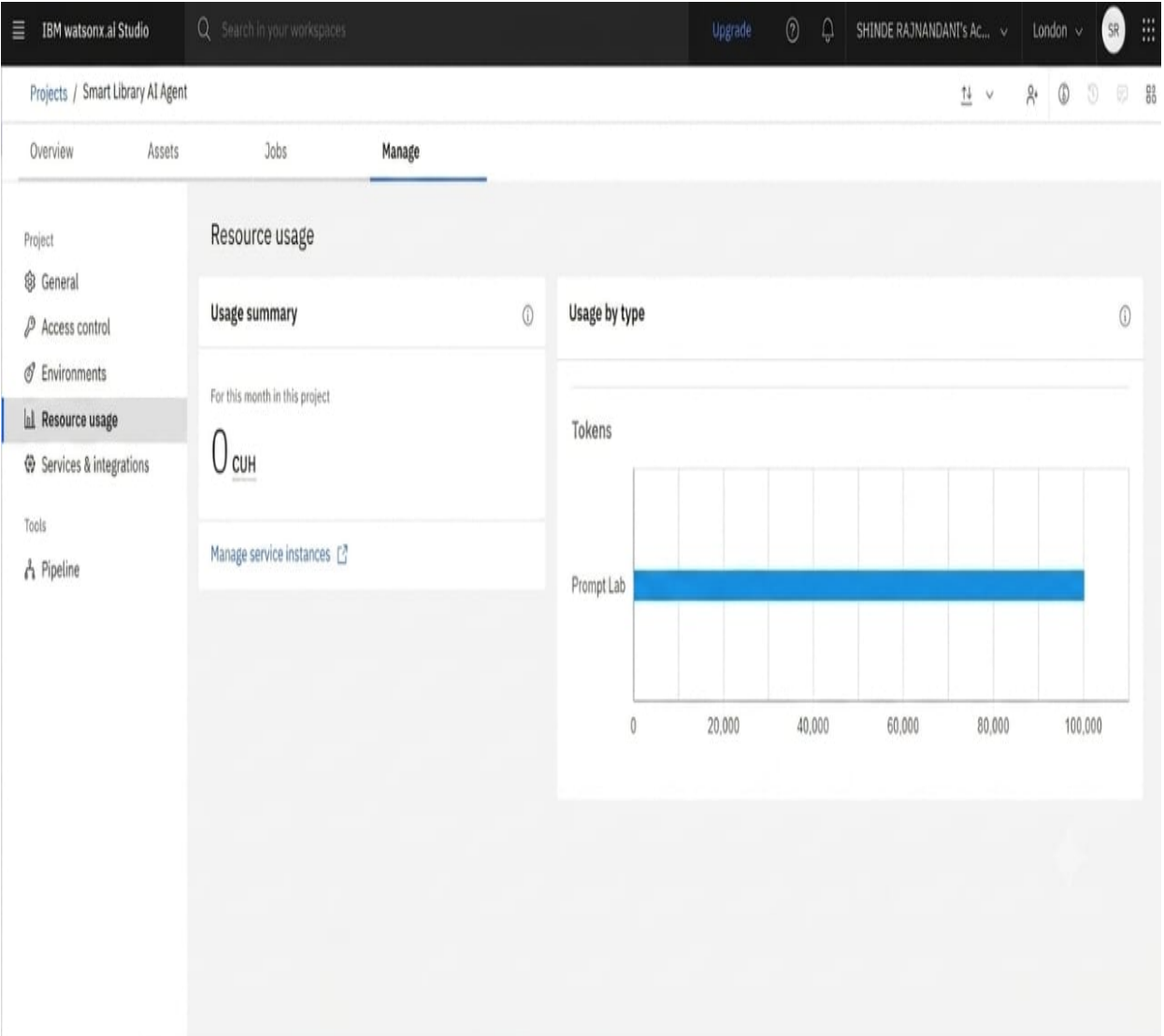


Token Usage

Before prompt



After prompt



[Projects](#) / Smart Library AI Agent

Overview

Assets

Jobs

Manage

Project

 **General** Access control Environments Resource usage Services & integrations

Tools

 Pipeline

Details



Name

Smart Library AI Agent

Description

An AI-powered Library Agent that recommends books, checks availability, assists with reservations, provides personalized learning resources, and answers student queries using IBM Granite and Watsonx Orchestrate.

Tags



Add tags to make projects easier to find.

Project ID

97185e1e-5c89-4076-bdcb-28a3449965f1



IBM Cloud

SHINDE RAJNAN...

SHINDE RAJNANDANI's Account

Filter by API key name or description

Create +

ID	Name	Description	Date created	Status	Enabled	Expires in	
	cpd-apikey-IBMid-6A7000G9N6-2026-06-18T06:01:16Z	API key created/managed by task credentials. It is managed for your use with Watson Studio operations. Please do not delete here.	6-18-2026 06:01 GMT		Yes	-	
	cpd-apikey-IBMid-6A7000G9N6-2026-07-05T14:36:37Z	API key created/managed by task credentials. It is managed for your use with Watson Studio operations. Please do not delete here.	7-5-2026 14:36 GMT		Yes	-	
	Smart Library AI Agent		7-8-2026 07:46 GMT		Yes	-	

1-25 items

IAM

Overview

Dashboard

Manage identities

Users

Trusted profiles

Service IDs

API keys

Identity providers

Manage access

Access groups

Authorizations

Roles

Novelty and Uniqueness :

The proposed **Library AI Agent** combines **IBM Granite AI**, **IBM watsonx.ai**, and **RAG (Retrieval-Augmented Generation)** to provide an intelligent and personalized library assistant for students.

- AI-Powered Book Recommendations** – Recommends books based on the student's subject, semester, course, or exam preparation.
- RAG-Based Smart Search** – Retrieves relevant books, syllabi, reading lists, and department guides before generating accurate responses.
- Natural Language Chatbot** – Understands questions asked in simple language and provides quick, relevant answers.
- Personalized Learning Support** – Helps students find suitable study materials based on their academic needs.
- Simple & User-Friendly Interface** – Allows students to search books and library resources through an easy-to-use chatbot.
- Scalable & Reliable** – Built using **IBM watsonx.ai**, **IBM Granite Model**, **IBM Cloud**, and **ChromaDB**, making it easy to expand with more books and departments.

Github Link :

Library AI Agent is an AI-powered chatbot that helps students quickly find the right books and study materials. It understands questions asked in simple language and provides book recommendations, syllabus information, reading lists, and library resources using **IBM watsonx.ai**, **IBM Granite**, and **RAG technology**. The project makes it easier and faster for students to access the information they need for their studies.

GitHub Repository:

<https://github.com/Rajnandani282/Smart-AI-Library-Agent>

Screenshots of project-

The screenshot displays the Grantha Library AI chat interface. The top header includes the Grantha Library AI logo, a moon icon for theme toggling, and a 'My Profile' button. A left sidebar contains navigation links: Chat, Search, For You, Trending, and My Books. The main chat area is titled 'AI Library Assistant' and includes a language selector set to 'English' and a trash icon. The chat history shows a previous message from the assistant (G) regarding book availability on shelf CS-D5, with a tip to use the search panel. Below this, two topic tags are visible: 'Database System Concepts' and 'Data Science from Scratch'. The current message from the user (S) asks, 'What subjects are there in the 5th semester CSE syllabus?'. The assistant's response (G) provides a detailed list of units and recommended books for the 5th semester CSE syllabus, including 'UNIT 1: Introduction & Physical Layer' through 'UNIT 5: Application Layer', all recommending 'Computer Networks (Tanenbaum)'. The assistant also advises contacting the department office or visiting the reading lists section for the full syllabus. The chat input field at the bottom shows the placeholder text 'Ask about books, topics, courses... (English / हिंदी / मराठी)' and a send button.

Grantha Library AI

Chat

Search

For You

Trending

My Books

AI Library Assistant

English

Shelf: CS-D5

Tip: Available copies are ready for immediate borrowing. Use the 'Search' panel to check live availability and reserve copies.

Database System Concepts Data Science from Scratch

08:36 PM

What subjects are there in the 5th semester CSE syllabus?

08:37 PM

Books and syllabus for 5th semester:

UNIT 1: Introduction & Physical Layer
Recommended Books: Computer Networks (Tanenbaum), Data Communications and Networking (Forouzan)

UNIT 2: Data Link Layer
Recommended Books: Computer Networks (Tanenbaum)

UNIT 3: Network Layer
Recommended Books: Computer Networks (Tanenbaum)

UNIT 3: Network Layer
Recommended Books: Computer Networks (Tanenbaum)

UNIT 4: Transport Layer
Recommended Books: Computer Networks (Tanenbaum)

UNIT 5: Application Layer
Recommended Books: Computer Networks (Tanenbaum)

For the full syllabus, contact your department office or visit the reading lists section at the library desk.

08:37 PM

Ask about books, topics, courses... (English / हिंदी / मराठी)

Screenshots of project-

The screenshot displays the Grantha Library AI web application. A modal window titled "Student Profile" is open, allowing users to create or update their profile. The background interface includes a top navigation bar with the Grantha logo, a search bar, and a "My Profile" button. A left sidebar contains links for Chat, Search, For You, Trending, and My Books. The main content area is currently empty, with a chat input field at the bottom.

Student Profile

Your Name
e.g. Arjun Sharma

Student ID
e.g. STU2024001

Department / Branch
Computer Science

Semester
— select —

Enrolled Courses (comma-separated)
e.g. Machine Learning, Data Structures, AI

Preferred Language
English

Cancel Save Profile

Ask about books, topics, courses... (English / हिंदी / मराठी)

Screenshots of project-

The screenshot displays the Grantha Library AI web application. A modal window titled "Student Profile" is open, allowing a user to enter their details. The background interface includes a sidebar with navigation options like "Chat", "Search", "For You", "Trending", and "My Books". At the top right, there are icons for a dark theme and a "My Profile" button. The bottom of the screen features a chat input area with a placeholder text and a send button.

Grantha Library AI

AI Library Assistant

English

Student Profile

Your Name

Rajnandani Shinde

Student ID

e.g. STU2024001

Department / Branch **Semester**

Computer Science 5th

Enrolled Courses (comma-separated)

data structure

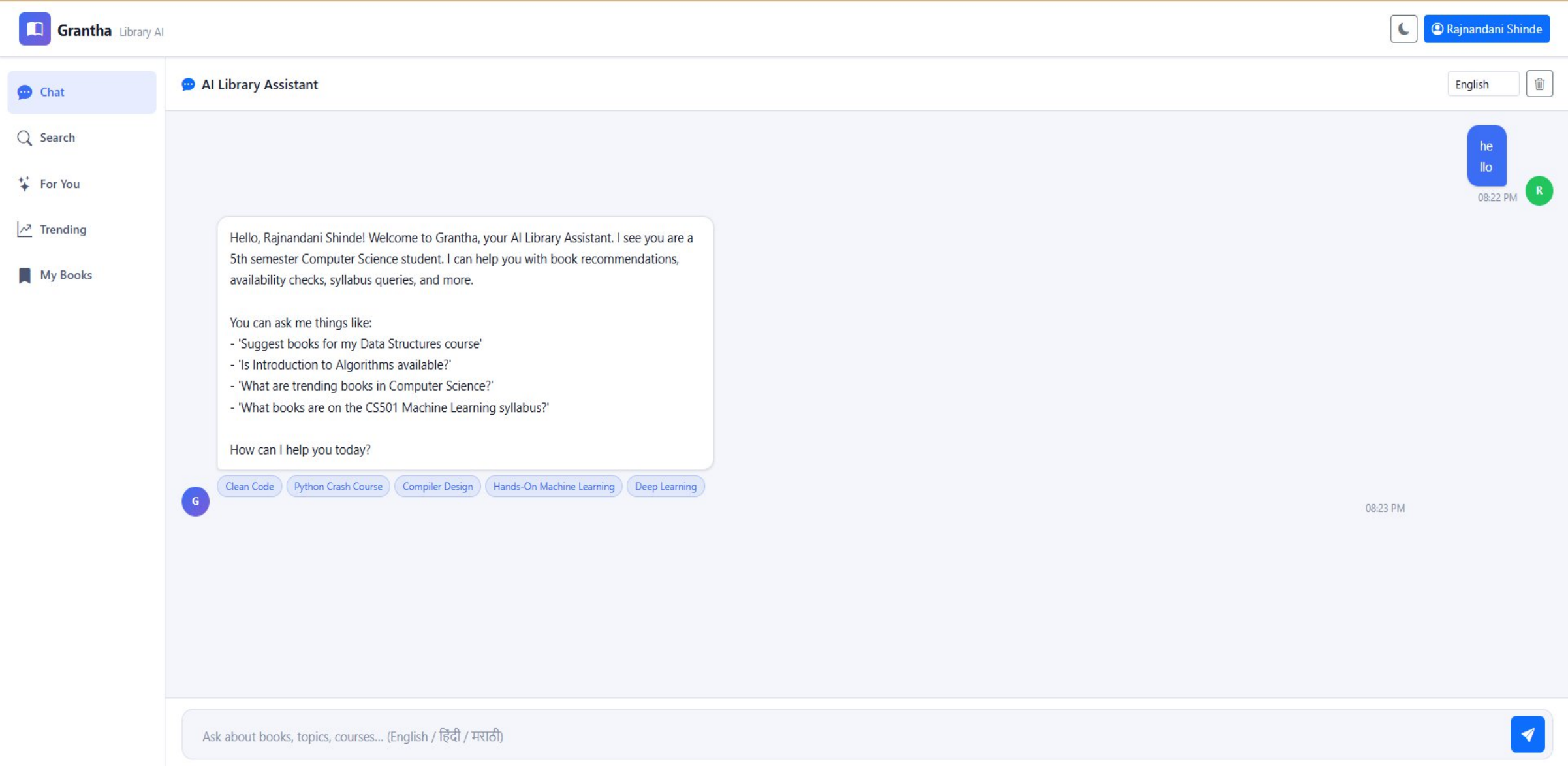
Preferred Language

English

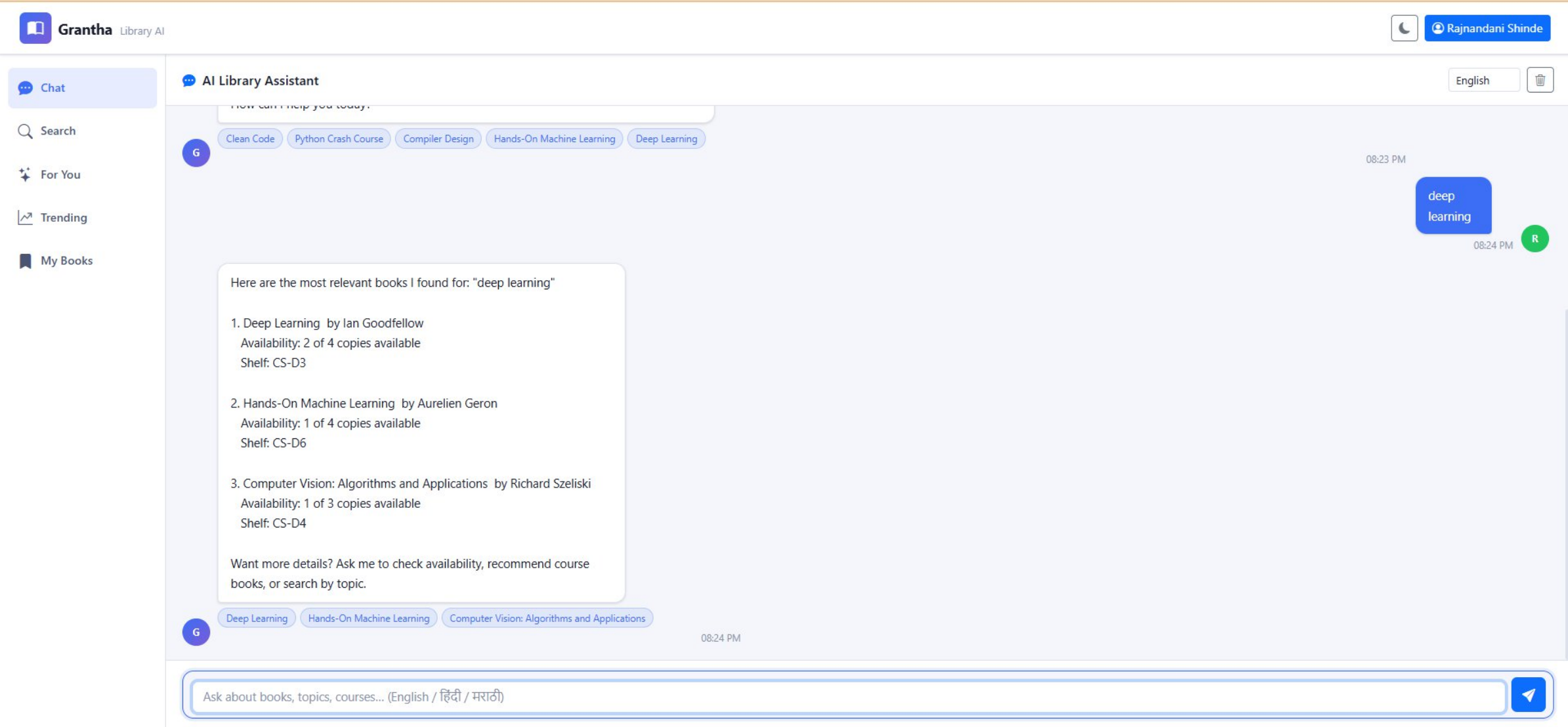
Cancel Save Profile

Ask about books, topics, courses... (English / हिंदी / मराठी)

Screenshots of project-



Screenshots of project-



Screenshots of project-

Personalised Recommendations

Refresh

Based on your profile: **Computer Science** **5th Sem** **data structure** [Edit profile](#)



Introduction to Algorithms

Thomas H. Cormen

Available (2/5)

CS-A1

Computer Science ·
2022

Relevant to data struct...



Data Structures and Algorithm

Analysis in C++

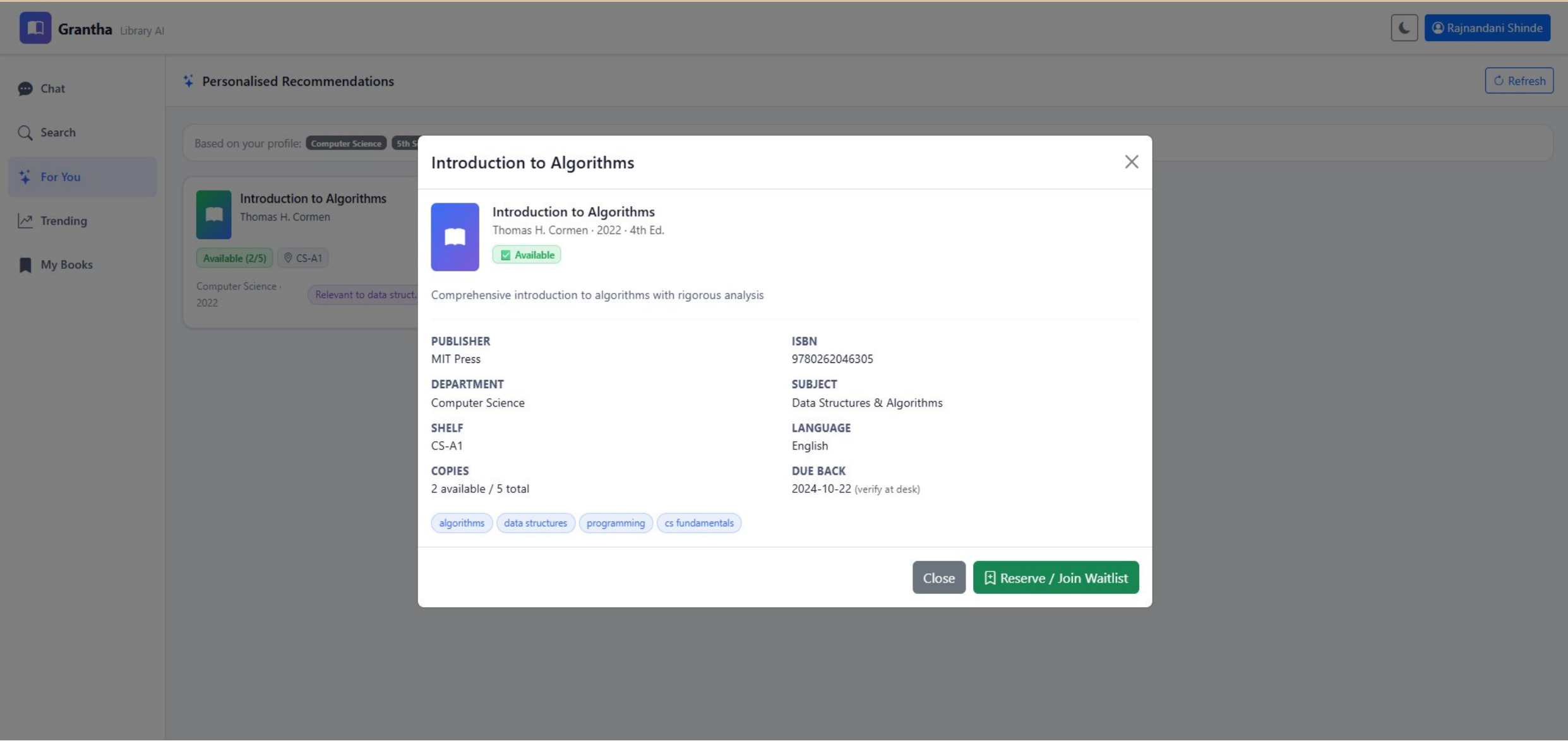
Available (1/4)

CS-A2

Computer Science ·
2014

Relevant to data struct...

Screenshots of project-



Screenshots of project-

 **Grantha** Library AI

Chat

Search

For You

Trending

My Books

Rajnandani Shinde

Refresh

Personalised Recommendations

Based on your profile: Computer Science 5th Sem data structure [Edit profile](#)

Introduction to Algorithms

Thomas H. Cormen

Available (2/5)

CS-A1

Computer Science · 2022

Relevant to data struct...

Data Structures and Algorithm Analysis in C++

Mark Allen Weiss

Available (1/4)

CS-A2

Computer Science · 2014

Relevant to data struct...

Reservation placed! Queue position: 1. The book is ready for pickup.

X

Screenshots of project-

Grantha Library AI

Chat

Search

For You

Trending

My Books

High-Demand & Trending Titles

All Departments

High demand (30+ borrows/semester)

Active waitlist

Available now

1

Calculus

James Stewart

Available (7/10)

MATH-A1

Mathematics

52 borrows/sem

2

Introduction to Algorithms

Thomas H. Cormen

Available (2/5)

CS-A1

Computer Science

47 borrows/sem

3 waiting

3

University Physics

Hugh D. Young

Available (6/8)

PHY-A1

Physics

44 borrows/sem

4

Hands-On Machine Learning

Aurélien Géron

Available (1/4)

CS-D6

Computer Science

41 borrows/sem

4 waiting

5

Artificial Intelligence: A Modern Approach

Stuart Russell

Available (1/3)

CS-D1

Computer Science

38 borrows/sem

5 waiting

6

Deep Learning

Ian Goodfellow

Available (2/4)

CS-D3

Computer Science

35 borrows/sem

2 waiting

7

Linear Algebra and Its Applications

David C. Lay

Available (4/6)

MATH-B1

Mathematics

33 borrows/sem

8

Operating System Concepts

Abraham Silberschatz

Available (3/6)

CS-B1

Computer Science

29 borrows/sem

1 waiting

Screenshots of project-

 Grantha Library AI

Chat

Search

For You

Trending

My Books

My Books & Waitlist

Refresh

Currently Borrowed

Holds & Waitlist

 Artificial Intelligence: A Modern Approach
Reserved: 2026-07-08 · Status:  Ready for pickup

Queue #1

 Calculus
Reserved: 2026-07-08 · Status:  Ready for pickup

Queue #1

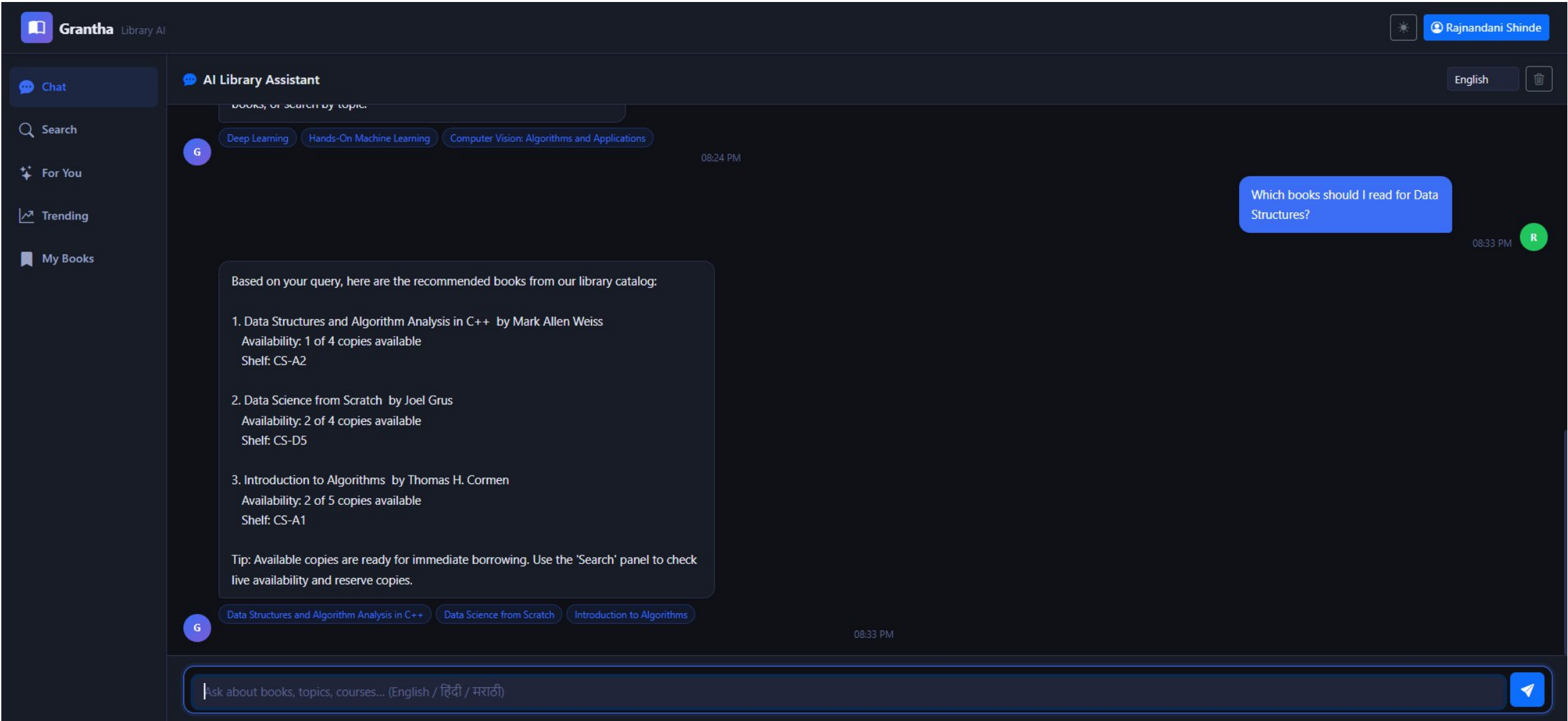
 Introduction to Algorithms
Reserved: 2026-07-11 · Status:  Ready for pickup

Queue #1

 Data Structures and Algorithm Analysis in C++
Reserved: 2026-07-11 · Status:  Ready for pickup

Queue #1

Screenshots of project-



Screenshots of project-

Grantha Library AI

English

Rajnandani Shinde

Chat

Search

For You

Trending

My Books

AI Library Assistant

2. Data Science from Scratch by Joel Grus
Availability: 2 of 4 copies available
Shelf: CS-D5

3. Introduction to Algorithms by Thomas H. Cormen
Availability: 2 of 5 copies available
Shelf: CS-A1

Tip: Available copies are ready for immediate borrowing. Use the 'Search' panel to check live availability and reserve copies.

G

Data Structures and Algorithm Analysis in C++

Data Science from Scratch

Introduction to Algorithms

08:33 PM

Is "Introduction to Algorithms" by Thomas H. Cormen available in the library?

R

08:34 PM

G

Introduction to Algorithms

08:34 PM

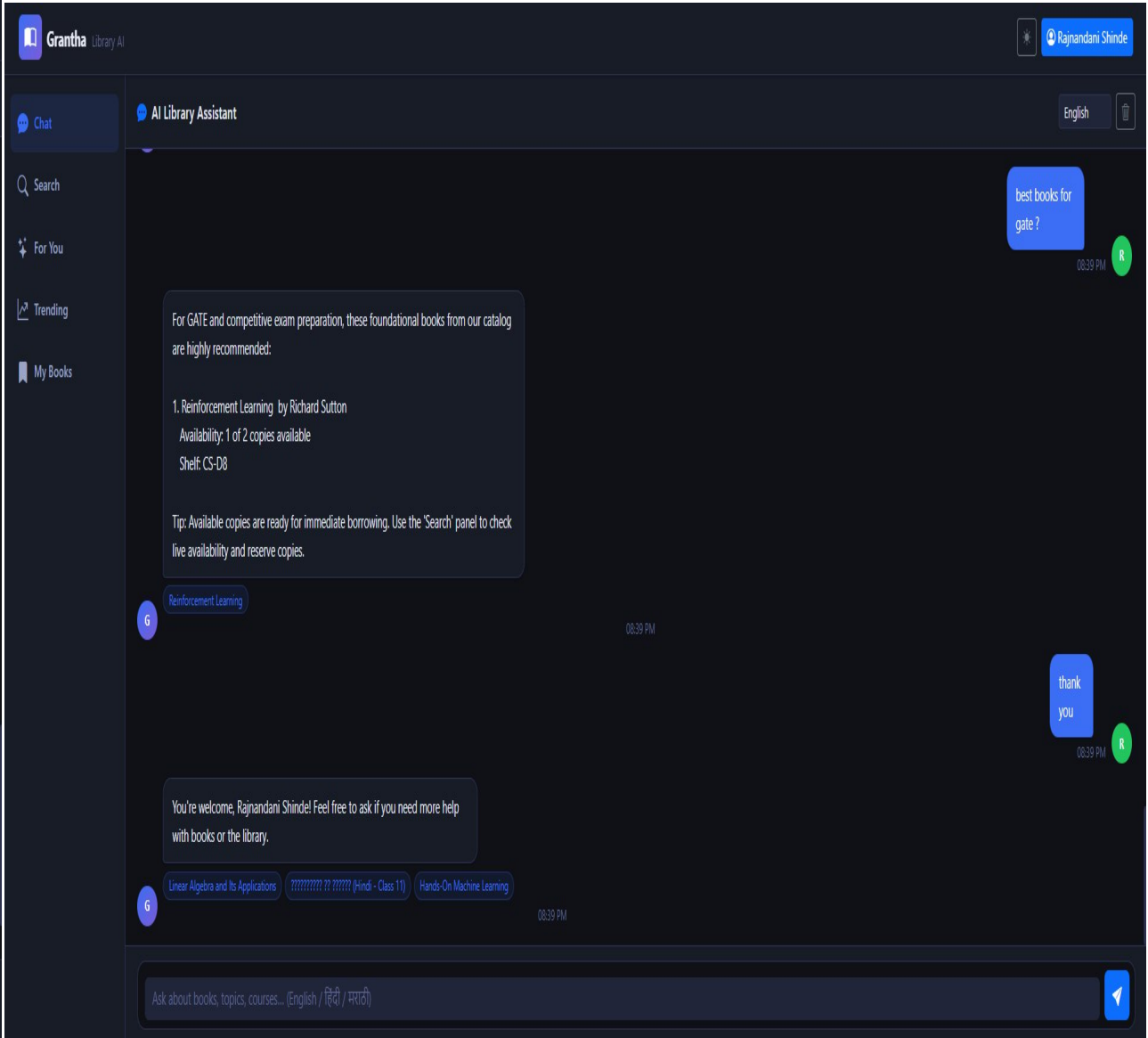
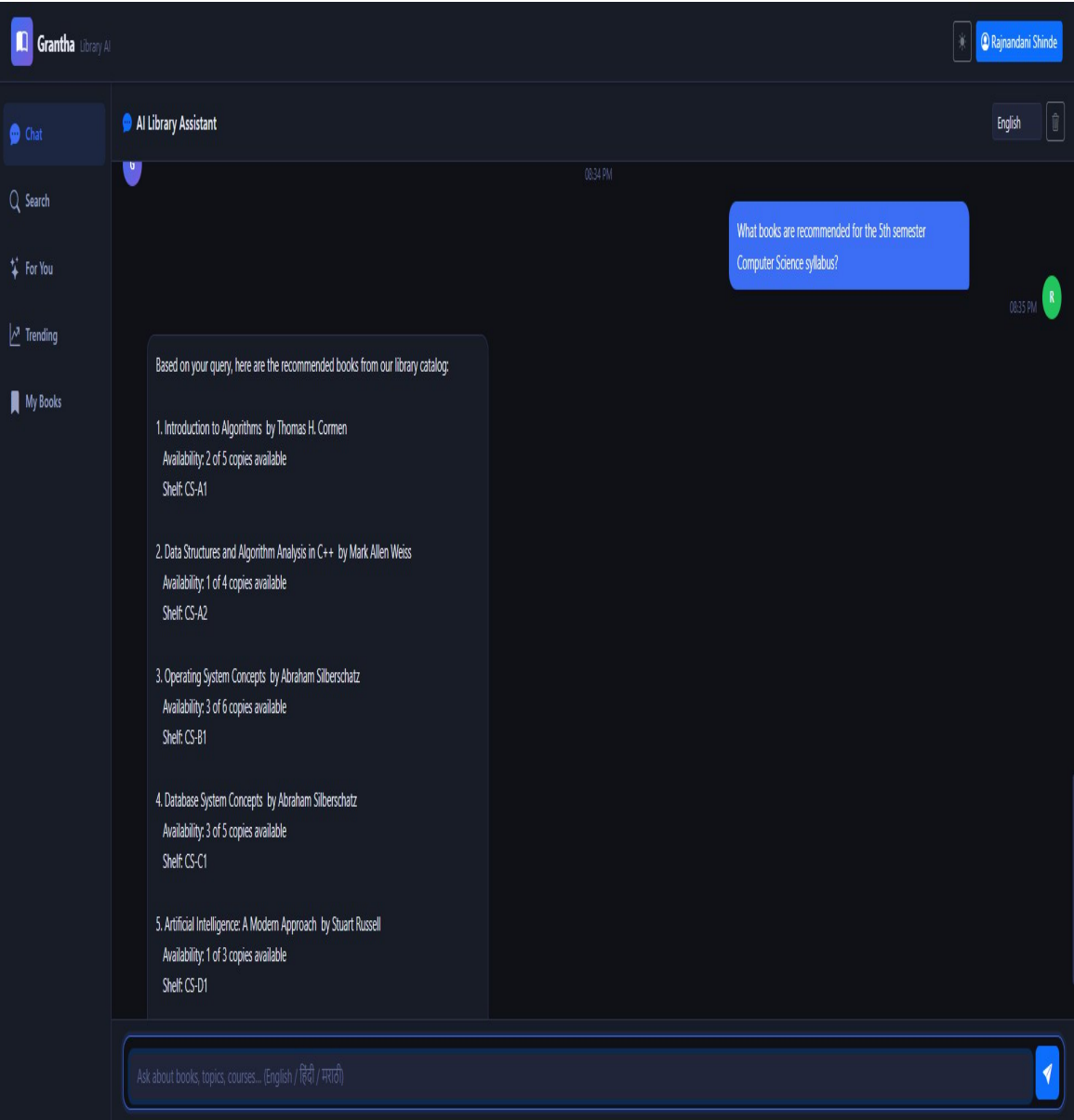
Here is the current availability for relevant books:

- Introduction to Algorithms
Status: AVAILABLE | 2 of 5 copies available | Shelf: CS-A1

Note: Please confirm exact availability at the circulation desk, as copies may have been issued or returned since last update.

Ask about books, topics, courses... (English / हिंदी / मराठी)

Screenshots of project-



The screenshot displays the Visual Studio Code interface for a project named "Smart AI Library Agent".

Explorer (Left Sidebar):

- Project: SMART AI LIBRARY AGENT
 - library_agent
 - __pycache__
 - knowledge_base
 - static
 - templates
 - vector_store
 - .env** (Selected)
 - .env.example
 - agent_instructions.py
 - app.py
 - rag_pipeline.py
 - README.md
 - requirements.txt
 - OUTLINE
 - TIMELINE

Main Editor (Center):

File: .env

```
library_agent > .env
1 # _____
2 # IBM watsonx.ai credentials - copy this file to .env and fill
3 # _____
4
5 # IBM Cloud API Key (IAM key from cloud.ibm.com → Manage → Access → API
6 IBM_API_KEY=cmjvDxbQBUJt_eM7x2t1YWaGfu1LuHOMVR3lDegFrKhx
7
8 # watsonx.ai Project ID (from watsonx.ai → Projects → <your project> →
9 WATSONX_PROJECT_ID=97185e1e-5c89-4076-bdcb-28a3449965f1
10
11 # watsonx.ai URL (region-specific endpoint)
12 WATSONX_URL=https://eu-gb.dataplatform.cloud.ibm.com
13
14 # _____
15 # Application settings
16 # _____
17
18 # Flask secret key (generate with: python -c "import secrets; print(secrets
19 FLASK_SECRET_KEY=change_this_to_a_random_secret_key
20
21 # Set to "development" for debug mode, "production" for deployment
22 FLASK_ENV=development
23
```

Terminal (Bottom):

PS C:\Users\DELL\OneDrive\Desktop\Smart AI Library Agent>

The screenshot displays the Visual Studio Code (VS Code) interface for a project named "Smart AI Library Agent".

Explorer Panel (Left): Shows the project structure. The "library_agent" folder is expanded, revealing subfolders like "_pycache_", "knowledge_base", "static", "templates", and "vector_store". Files listed include ".env", ".env.example", "agent_instructions.py", "app.py", "rag_pipeline.py", "README.md", and "requirements.txt".

Main Editor: Displays the content of the ".env" file. The configuration includes:

- Vector store backend: "chroma" or "faiss" (set to "chroma").
- Embedding model: sentence-transformers/all-MiniLM-L6-v2.
- LLM model ID: ibm/granite-13b-chat-v2.
- ChromaDB persistence directory: ./vector_store/chroma_db.
- FAISS index path: ./vector_store/faiss_index.
- Library name: University Central Library.
- Maximum chunks retrieved per RAG query: 6.

Terminal Panel (Bottom): Shows the PowerShell prompt at the project root: "PS C:\Users\DELL\OneDrive\Desktop\Smart AI Library Agent>".

Screenrecording of project-

https://drive.google.com/file/d/100moZ0zNA05Cyni8tgupBi8ms3Zel4xw/view?usp=drive_link

Certificates Earned :

IBM SkillsBuild	Completion Certificate
	This certificate is presented to SHINDE RAJNANDANI for the completion of Exploring Quantum Computing (ALM-COURSE_3825253) According to the Adobe Learning Manager system of record
Completion date: 29 Jun 2026 (GMT)	Learning hours: 50 mins

IBM SkillsBuild	Completion Certificate
	This certificate is presented to SHINDE RAJNANDANI for the completion of Level Up Cybersecurity with Generative AI (ALM-COURSE_3947352) According to the Adobe Learning Manager system of record
Completion date: 29 Jun 2026 (GMT)	Learning hours: 1 hr 30 mins

In recognition of the commitment to achieve
professional excellence



Rajnandani Shinde

Has successfully satisfied the requirements for:

Getting Started with Cybersecurity

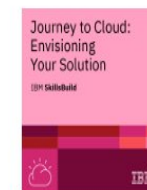


Issued on: Jun 24, 2026
Issued by: IBM SkillsBuild

Verify: <https://www.credly.com/badges/e879ee0b-3e0a-4f52-8d69-7ed6518f5bc5>



In recognition of the commitment to achieve
professional excellence



Rajnandani Shinde

Has successfully satisfied the requirements for:

Journey to Cloud: Envisioning Your Solution



Issued on: Jun 20, 2026
Issued by: IBM SkillsBuild

Verify: <https://www.credly.com/badges/a953654a-9928-4650-a945-2e903f7e13fc>



IBM SkillsBuild

Completion Certificate



This certificate is presented to
SHINDE RAJNANDANI

for the completion of

Lab: Troubleshoot Your Code Using IBM Bob

(ALM-COURSE_4071307)

According to the Adobe Learning Manager system of record

Completion date: 20 Jun 2026 (GMT)

Learning hours: 30 mins

In recognition of the commitment to achieve
professional excellence



Rajnandani Shinde

Has successfully satisfied the requirements for:

Getting Started with Artificial Intelligence



Issued on: Jun 20, 2026

Issued by: IBM SkillsBuild

Verify: <https://www.credly.com/badges/646a7ed1-da76-4c08-9802-44cc6bda1e75>



Future Scope

- Integrate with a **real-time library database** to show live book availability.
- Add **book reservation and waitlist** features.
- Support **multiple languages** such as English, Hindi, and Marathi.
- Develop a **mobile application** for Android and iOS.
- Add **voice-based search** for hands-free interaction.
- Provide **personalized book recommendations** based on students' interests and learning history.
- Integrate with the **college ERP/Library Management System** for student login and borrowing history.
- Expand the knowledge base by adding **more books, journals, research papers, and digital resources**.

Future Goal:

Transform the system into a **smart AI-powered digital library assistant** that provides personalized learning support, quick access to library resources, and an enhanced learning experience for students.

Thank You!

Thank you for your time and interest.